AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM112
Introduction to Computing and Information Technology

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 1.5 hours

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple Choice Questions(5 marks):

1) To code 15 colors, we need
   a. 16 bits
   b. 15 bits
   c. 4 bits
   d. 5 bits
2) Which of the following is true for the sign-magnitude representation?
   a. There are 2 different encodings for 0.
   b. If the leftmost bit of the number is 1, then the number is negative.
   c. In an 8-bit format, only 7 bits are used to represent the magnitude.
   d. All the above
3) The equivalent decimal number of the signed binary number 10001111 is:
   a. +15
   b. -15
   c. +143
   d. -143
4) ----------in the context of UI design means making it clear what a UI element is used for.
   a. Visibility
   b. Tolerance
   c. affordance.
   d. simplicity.
5) A----------represents the time between adjacent peaks (or troughs).
   a. cycle
   b. frequency
   c. amplitude
   d. None of the above

Part 2: Short questions(10 marks):

Question 1: (3 marks)

How you can protect your computer against malware? List three points only

Question 2: (4marks)

For the given numbers find their equivalent in the specified base. **Show your steps.**

$(111000)_2 = ()_8$

$(1A)_{16} = ()_{10}$

$(672)_8 = ()_{16}$

$(0.75*4)_{10} = ()_2$

Question 3: (3 Marks)

Bitmap files are relatively large for representating an image. Mention and describe two techniques for representing an image

Part 3: Problem-solving / Long Questions (15 marks)

Question 1:(6 marks)

Consider an image of size 200 x 120. How many bits and bytes are required to store this image inside the computer

a. Using only 2 colors: black and white.

b. Using 4 bits per pixel in a grayscale model.

c. Using RGB model.

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Question 2: (4 marks)
List 4 Usability principles and give the definition of each of them?

Question 3: (5 marks)

a) Difference between Structured and Unstructured data

b) List any 3 important rolesthat are required in a Data Science Project and explain brefly

End of Questions

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